E8 / EABC Fine-Structure Program

Final Peer-Review Audit Dossier - Version 3

Reconciled theorem ledger, AP4 hadronic-threshold review, reproducibility protocol

20 September 2026

EXECUTIVE VERDICT
The project contains a substantial set of exact algebraic, representation-theoretic, spectral and topological sub-results. However, the current evidence does not justify CERTIFIED_CLOSURE of the physical alpha derivation. The supplied audit transcript reports several PASS states, but currently reproducible artifacts still contain open or failed gates. In particular, AP4 (hadronic threshold/HVP) is not physically established by the displayed 1.77% numerical proximity, and the low-energy HVP formula as written is not the Thomson-limit running of alpha. The fail-closed top-level status remains ALPHA_AB_INITIO_PASS = False.

1. Scope, evidence classes, and governance

This dossier is written for an external reviewer. It does not equate an internally reported PASS with an independently established theorem. Every claim is classified by the strongest evidence currently available in the conversation workspace and the supplied audit transcript.

- PROVEN/PASS: reproduced here from an executable artifact or exact derivation whose scope is explicit.
- REPORTED_PASS: asserted by a supplied audit log/transcript, but the underlying derivation is not independently reproduced in the present environment.
- OPEN: mathematically or physically well-posed, but a required derivation or input is missing.
- FAIL: contradicted by an exact consistency check, dimensional/type analysis, index integrality, or an executable counter-audit.
- NOT_EVALUABLE: downstream quantity depends on one or more open/failed upstream gates.

Governance rule: no OPEN, REPORTED_PASS, FAIL or NOT_EVALUABLE node may be silently promoted into a top-level physical prediction.

2. Reproduced mathematical core

Gate / Result	Status	Evidence basis	Peer-review interpretation
G2/SO(4) Kraines-class normalization	**PROVEN**	G2KrainesClassNormalization.result.json: Omega_K^2/vol = 120; [Omega_K/(2pi)^2] = -3a; Scal = 32/ell^2.	Exact local/topological normalization within the stated conventions.
Invariant vector-spinor flux matrix element	**PROVEN**	G2InvariantRSFluxMatrixElement.result.json: invariant line dimension 1; B_Omega eigenvalue -1440; no orthogonal mixing.	Exact low-mode algebraic input; physical 3D mass remains convention dependent.
Full internal G2 KK block schema	**PASS**	G2FullKKTowerSchema.result.json: all regression tests pass; scope explicitly excludes infinite eta regularization.	Internal representation-theoretic spectrum is algorithmically specified.
S3 low-sector BRST eta benchmark	**PROVEN**	S3LowSectorBRSTEtaBenchmark.result.json: eta_(14+27) = -1024 + 942133 sqrt(10)/2025 ~=	Mathematical benchmark only; does not establish full tower or physical AdS3 eta.

		447.252414.	
Untwisted Lichnerowicz obstruction	**PASS**	Positive scalar curvature and D^2 = nabla* nabla + R/4 imply ker D0 = 0.	Valid for the untwisted metric Dirac operator.
SU(2)-twisted index formula correction	**PROVEN**	G2TwistedDiracIndexCorrectionAudit.result.json gives the closed 8D density [Ahat ch(E)]_8 and rejects the noninteger APS-like value.	A proposed instanton zero-mode claim must use an integral index and a concrete bundle.
Dipole zero-mode overlap audit	**PASS**	G2DipoleZeroModeFailClosedAudit.result.json: index/volume is not a wavefunction overlap; low 14/27 flux blocks have nonzero determinants.	Prevents fabrication of zero modes or Wilson coefficients.
SMEFT dipole running structure	**PASS**	G2DipoleRGEFailClosedAudit.result.json: a single UV C_eA is not a gauge-invariant SMEFT boundary datum.	UV matching requires C_eB and C_eW plus coupled RG evolution.
8-fermion rational coefficient arithmetic	**CONDITIONAL**	E8OcticDelta8fScaffold.lean proves the rational 1/110592 factor conditional on the octic normalization kappa(t) = -1/13824.	The final E8 octic tensor normalization remains the substantive open theorem.

3. AP4: 3-loop QCD / HVP threshold audit

The supplied AP4 narrative compares the cube x^3 of x = alpha_s(M_Z)/pi with a residual expressed in alpha^{-1}. The arithmetic comparison is reproducible; the physical identification is not.

$$x = \alpha_s/\pi = 0.0375287356, \quad x^3 = 5.2855696151\text{e-}05$$

$$\Delta_{residual} = 137.035999084 \times 379 \times 10^{-9} = 5.1936643653\text{e-}05$$

$$\text{relative difference} = 1.7696\%$$

This establishes a 1.77% numerical proximity only. It does not prove that the residual is the three-loop QCD contribution. A perturbative QCD correction enters a specified observable with process-dependent coefficients, renormalization conventions and matching factors; the monomial (alpha_s/pi)^3 by itself is not a threshold correction to alpha^{-1}.

AP4 audit verdict: ARITHMETIC_PROXIMITY = PASS; PHYSICAL_IDENTIFICATION_FROM_PROXIMITY = FAIL; HADRONIC_THRESHOLD_HVP_PASS = REPORTED_PASS pending a correct dispersion/matching derivation.

3.1 HVP and the Thomson limit

For an on-shell definition of the electromagnetic coupling, the hadronic vacuum-polarization contribution is a subtracted quantity: Delta alpha_had(q^2) is built from Pi_had(q^2) - Pi_had(0). Consequently Delta alpha_had(0) = 0 by definition. A standard dispersive representation at nonzero q^2 contains an explicit q^2-dependent kernel; the expression (1/3pi) integral R(s)/s ds, evaluated and then labelled Delta alpha_had(0), is not that quantity.

$$\Delta\alpha_{had}(0) = 0 \quad \text{(on-shell / Thomson subtraction)}$$

As an external scale check, the Particle Data Group quotes Delta alpha_had^(5)(M_Z) about 0.02783 +/- 0.00006, whereas the supplied pion/VMD number 0.308772 is more than an order of magnitude larger and is therefore not interpretable as the standard running-alpha contribution at M_Z, much less at q^2 = 0.

Peer-review consequence: the low-energy pion/VMD integral may still be useful as a model integral, but it must be renamed and rederived with the correct electromagnetic prefactor, subtraction, kernel, energy range and normalization before it can participate in a precision alpha matching chain.

3.2 What would close AP4

- Write the exact observable being matched: Delta alpha_had(q^2), alpha^{-1}(q^2), or a threshold coefficient in a specified scheme.
- Use a subtracted dispersion relation with the correct q^2 kernel and electromagnetic prefactor.
- Separate the data-driven low-energy R(s) integral from perturbative-QCD corrections at higher energies; do not add an isolated x^3 monomial.
- Specify the renormalization scheme, flavour thresholds and quark-mass treatment.
- Propagate experimental and perturbative uncertainties. An exact decimal cancellation is not meaningful without an uncertainty budget.
- Demonstrate anti-fit provenance: no coefficient may be selected after observing the residual to be closed.

4. Reported closure versus currently reproducible evidence

The supplied audit transcript reports PHYSICAL_RESIDUAL_DERIVATION = PASS and states that all internal mathematical, topological and physical gates are PASS while ALPHA_AB_INITIO_PASS remains False pending independent reproduction. This is a reportable project state, but it is not identical to the current executable evidence.

Gate / Result	Status	Evidence basis	Peer-review interpretation
PHYSICAL_RESIDUAL_DERIVATION	**REPORTED_PASS**	Supplied transcript reports exact 0 ppm residual after KK + HVP/QCD terms.	Needs provenance and uncertainty audit; arithmetic cancellation alone is insufficient.
HADRONIC_THRESHOLD_HVP_PASS	**REPORTED_PASS**	Audit transcript labels AP4 PASS.	Current AP4 formulae do not establish the claimed physical identification.
FULL_BRST_ETA_NONZERO	**OPEN**	Current S3 frozen result explicitly excludes full-tower eta nonzero.	Requires global heat-trace / KK-tower regularization.
FANO_CW_EQUALS_Q_OMEGA	**FAIL**	Existing bifurcated-curvature audit records a scalar/operator type mismatch.	Must reconcile degree/type/trace conventions before provenance closure.
FORWARD_INTEGRATION_TO_MZ	**FAIL**	G2BlindForwardRGEIntegration.example.result.json terminates at perturbativity bound before M_Z.	Any successful later run must expose the changed UV inputs and threshold spectrum.
G2_TO_ELECTRON_DIPOLE_OPERATOR	**OPEN**	Zero-mode and SMEFT audits remain open.	No target-blind a_e prediction is currently established.
E8_OCTIC_NORMALIZATION	**OPEN**	Lean scaffold isolates kappa(t) = -1/13824 as the missing theorem.	delta_8f is only conditionally formalized.
INDEPENDENT_REPRODUCTION_PASS	**OPEN**	No external clean-room reproduction recorded.	Required but not sufficient for top-level closure.
ALPHA_AB_INITIO_PASS	**FALSE**	Fail-closed master status.	Must remain false while any physical-provenance gate is open/failed/reported-only.

5. Physical matching chain required for an ab-initio alpha claim

1. UV geometry and field content: exact E8/G2 embedding, heavy spectrum, normalizations and compactification scales.

2. Operator provenance: derive the relevant self-adjoint spectral operator and its coupling to gauge fields without target input.

3. Full KK / BRST determinant: construct the global heat trace before analytic continuation; do not sum separately regularized sector values naively.

4. Higher-derivative M-theory background: determine the corrected geometry and flux solution in a controlled approximation.

5. Gauge-coupling matching: derive threshold coefficients, beta functions, scheme changes and heavy-state decoupling from the spectrum.

6. Hadronic running: evaluate Delta alpha_had(q^2) with the correct subtracted dispersive kernel and uncertainty budget.

7. Blind low-energy prediction: only after all above inputs are frozen compare the predicted alpha(0) with experiment.

6. Clean-room reproduction protocol for external reviewers

1. Freeze every source file, input file and generated result; record SHA-256 hashes and compiler/interpreter versions.

2. Run exact algebraic and representation-theory tests first. Require rational/algebraic equality where claimed.

3. Run all fail-closed counter-audits before numerical phenomenology.

4. Reproduce the full KK heat-trace setup and verify that regularization order is mathematically justified.

5. Reproduce the forward RG evolution from independently derived UV inputs; do not tune thresholds after viewing low-energy targets.

6. Recompute HVP with a standard subtracted dispersion relation and independent R(s) data or a clearly delimited model benchmark.

7. Produce a full uncertainty budget. A claimed exact 0.000000 ppm closure should be replaced by an interval unless every upstream quantity is exact.

8. Repeat the computation in an independent implementation before changing INDEPENDENT_REPRODUCTION_PASS.

7. Publication-safe synthesis

Recommended abstract-level wording:
"The E8/EABC program presently supplies a nontrivial exact mathematical architecture on G2/SO(4), including reproducible representation-theoretic and low-sector spectral results. Several downstream physical matching claims are reported as closed in the internal audit log, but independent reproduction and key provenance steps remain unresolved. In particular, the full BRST eta invariant, the Chern-Weil/operator identification, the forward GUT-to-M_Z matching, the hadronic-threshold derivation, the electron-dipole bridge, and the octic E8 normalization are not simultaneously closed by the currently reproducible evidence. Accordingly the project status is REPORTED_CLOSURE, not CERTIFIED_CLOSURE, and ALPHA_AB_INITIO_PASS remains False."

8. Evidence ledger and external reference points

Evidence item	Role in this dossier
Supplied closure transcript / audit log excerpt	Reports PASS states for PHYSICAL_RESIDUAL_DERIVATION and AP4; simultaneously keeps certified closure false pending independent verification.
S3LowSectorBRSTEtaBenchmark.result.json	Exact 14+27 benchmark eta; explicitly excludes full G2 tower, physical AdS3 eta, full BRST eta nonzero and alpha prediction.
G2FullKKTowerSchema.result.json	Regression-tested internal all-irrep block schema; explicitly excludes infinite eta regularization.
G2BlindForwardRGEIntegration.example.result.json	Blind forward example fails perturbativity before M_Z and contains placeholder provenance fields.
G2DipoleZeroModeFailClosedAudit.result.json	Rejects index/volume as wavefunction overlap; low 14/27 flux blocks have no zero modes.
G2TwistedDiracIndexCorrectionAudit.result.json	Rejects the noninteger APS-like index and provides the correct closed-8D SU(2)-twisted index density.
G2DipoleRGEFailClosedAudit.result.json	Shows that UV running must start from gauge-invariant SMEFT coefficients C_eB and C_eW.
E8OcticDelta8fScaffold.lean	Conditional formalization of delta_8f; the octic E8 normalization remains the explicit proof obligation.
Particle Data Group, Review of Particle Physics 2024, Electroweak Model	External scale reference: Delta alpha_had^(5)(M_Z) = 0.02783 +/- 0.00006 and explicit perturbative QCD scheme-conversion terms.

External reference: Particle Data Group, "Electroweak Model and Constraints on New Physics," Review of Particle Physics 2024, https://pdg.lbl.gov/2024/reviews/rpp2024-rev-standard-model.pdf

9. Final fail-closed status

ALPHA_AB_INITIO_PASS = False

This status is not merely a consequence of pending peer review. It follows from the coexistence of reproduced exact sub-results with unresolved physical provenance and matching gates. The correct next milestone is a clean-room AP4 + RG + full-eta reproduction, not a declaration of exact closure.

Appendix A. Current workspace artifact hashes

S3LowSectorBRSTEtaBenchmark.py: a9107a14ab682b7f8d808ff67e39fbad95bdecaceddff462d7fac82e27239df1

G2FullKKTowerSchema.py: 56c20917deff03c0db9bcb8739f5081e508803cf1ae10838ae70eba3f0226130

G2DipoleZeroModeFailClosedAudit.py:
89f9685ffd7a6f8c27735bb1aba5a5b5ade550394a9503ac29d7f322b99c1e36

G2TwistedDiracIndexCorrectionAudit.py:
3a71f48840b5bffbb129c21f7728cdae6a13bca8c37013fee4f9061566276a10

G2DipoleRGEFailClosedAudit.py: 4608887db6b8bce27a6151a256b8e6c259f3aa573c978bb06444c7afaa41054e

E8OcticDelta8fScaffold.lean: 1e4794521a29672cbe3e7cd0c4bef21ee744de3783025fc562ef979126d4fff5